

9. Non-parametric statistics - Kruskal Wallis

statOmics, Ghent University (<https://statomics.github.io>)

Lieven Clement

Comparison of g groups

- Extend F -test from a one-way ANOVA to non-parametric alternatives.

DMH example

Assess genotoxicity of 1,2-dimethylhydrazine dihydrochloride (DMH) (EU directive)

- 24 rats
- four groups with daily DMH dose
 - control
 - low
 - medium
 - high
- Genotoxicity in liver using comet assay on 150 liver cells per rat.
- Are there differences in DNA damage due to DMH dose?

Comet Assay:

- Visualise DNA strand breaks
- Length comet tail is a proxy for strand breaks.

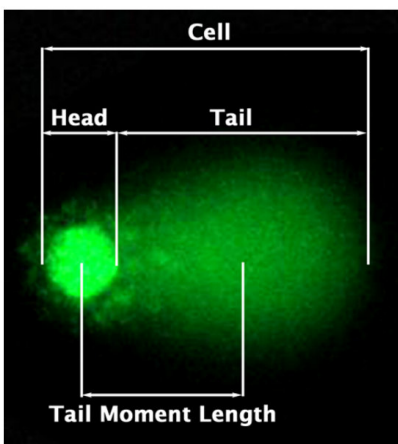
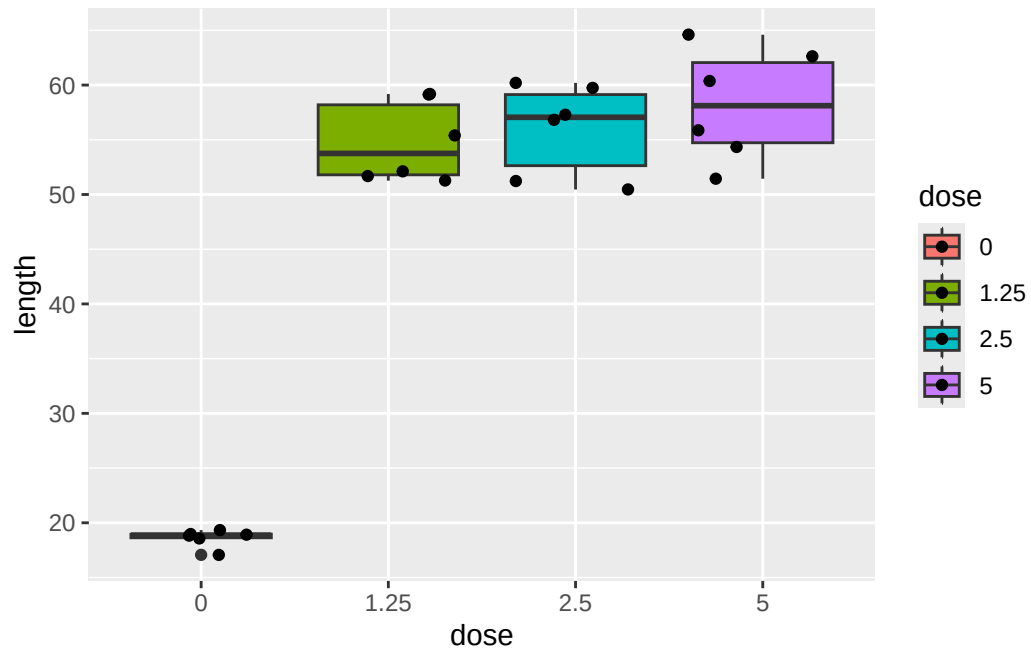


Figure 1: Comet assay

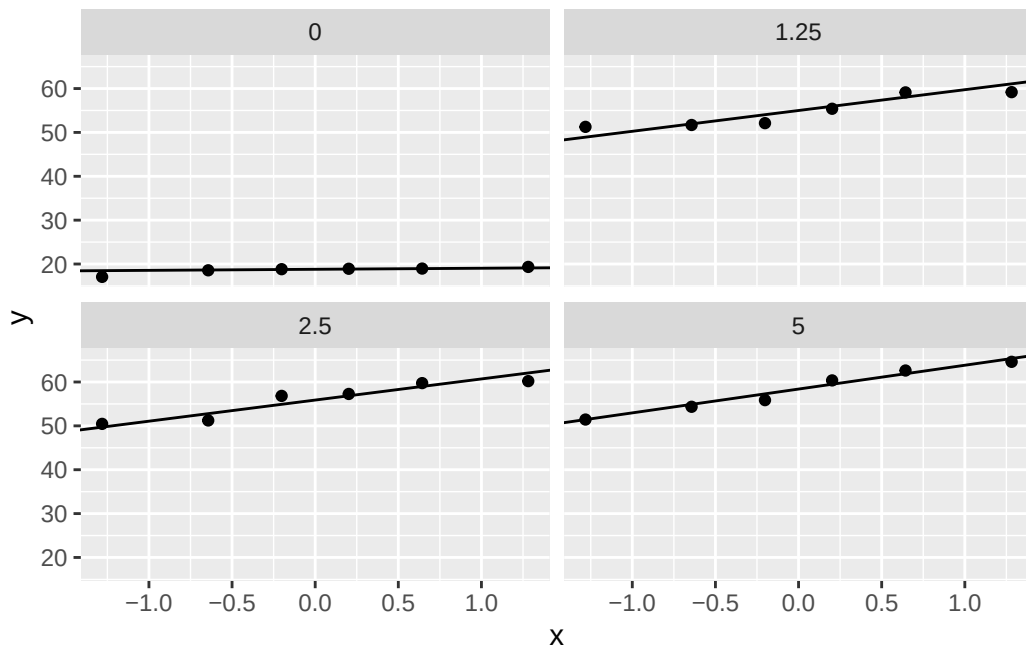
```
dna <- read_delim("https://raw.githubusercontent.com/GTPB/PSLS20/master/data/dna.txt", delim = ";")
dna$dose <- as.factor(dna$dose)
dna
```

```
# A tibble: 24 x 3
  id   length dose
<chr> <dbl> <fct>
1 Rat1    19.3 0
2 Rat2    18.9 0
3 Rat3    18.6 0
4 Rat4    19.0 0
5 Rat5    17.1 0
6 Rat6    18.8 0
7 Rat7    55.4 1.25
8 Rat8    59.2 1.25
9 Rat9    59.1 1.25
10 Rat10   52.1 1.25
# i 14 more rows
```

```
dna |>
  ggplot(aes(x = dose, y = length, fill = dose)) +
  geom_boxplot() +
  geom_point(position = "jitter")
```

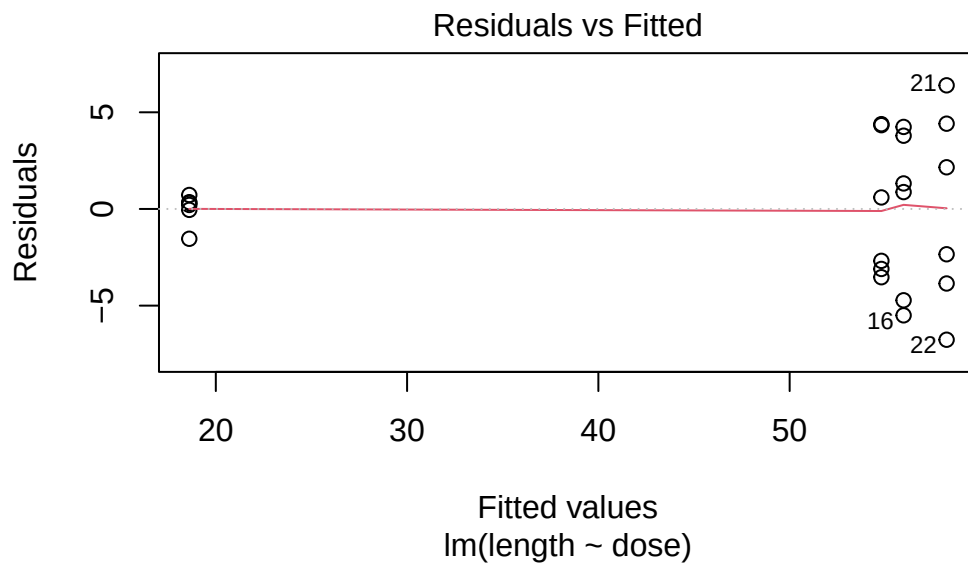


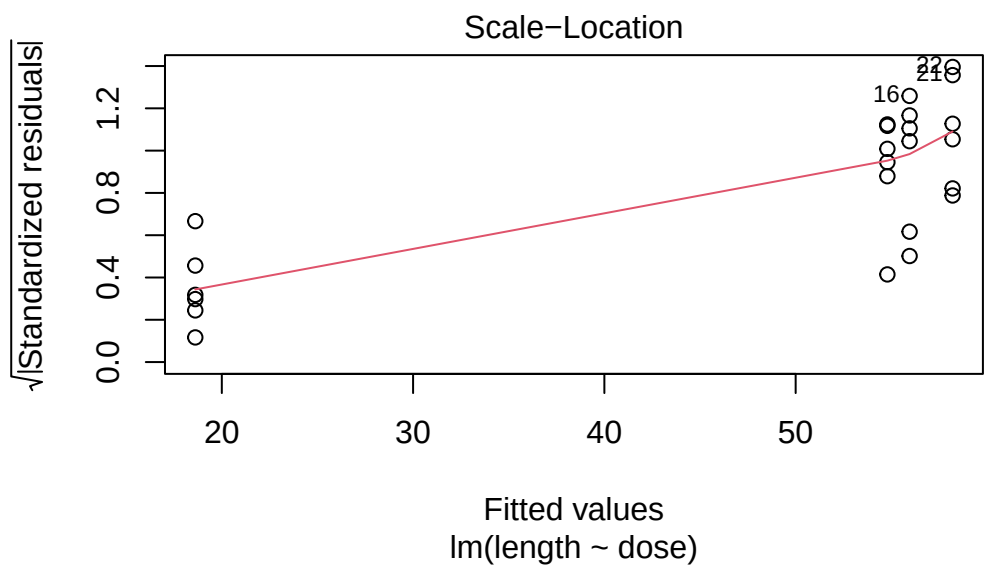
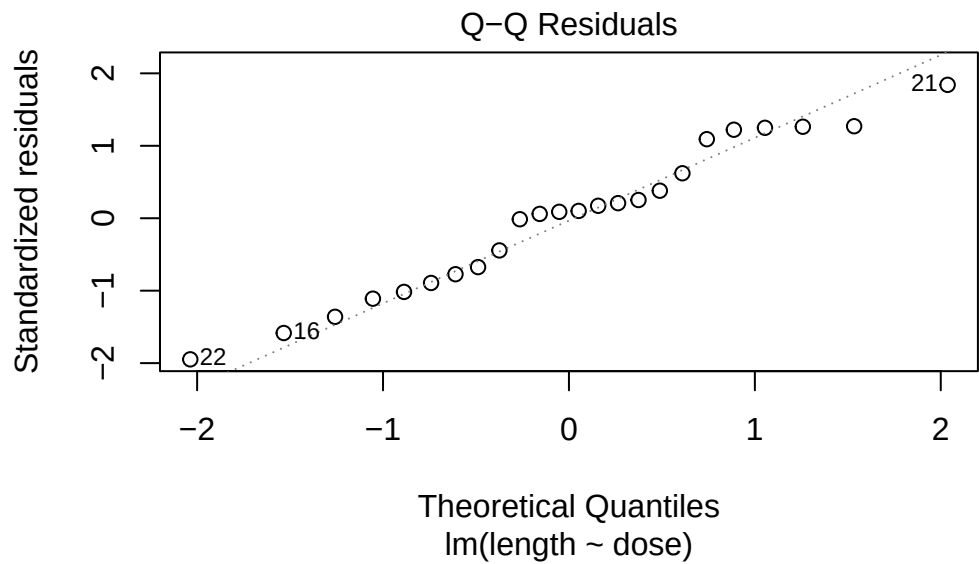
```
dna |>
  ggplot(aes(sample = length)) +
  geom_qq() +
  geom_qq_line() +
  facet_wrap(~dose)
```

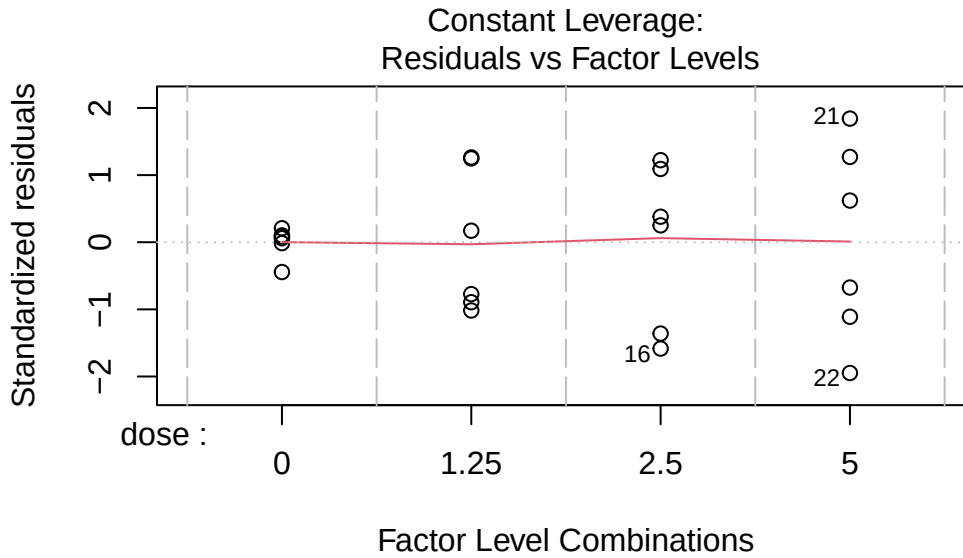


- Strong indication that data in control group has a lower variance.
- 6 observations per group are too few to check the assumptions

```
plot(lm(length ~ dose, data = dna))
```







Kruskal-Wallis Rank Test

- The Kruskal-Wallis Rank Test (KW-test) is a non-parametric alternative for ANOVA F-test.
- Classical F -test statistic can be written as

$$F = \frac{SST/(g-1)}{SSE/(n-g)} = \frac{SST/(g-1)}{(SSTot - SST)/(n-g)},$$

- with g the number of groups.
- SSTot depends only on outcomes $\mathbf{y}$ and will not vary in permutation test.
- SST can be used as statistic

$$SST = \sum_{j=1}^t n_j (\bar{Y}_j - \bar{Y})^2$$

- The KW test statistic is based on SST on rank-transformed outcomes¹,

$$SST = \sum_{j=1}^g n_j (\bar{R}_j - \bar{R})^2 = \sum_{j=1}^t n_j \left(\bar{R}_j - \frac{n+1}{2} \right)^2,$$

¹we assume that no *ties* are available

- with $\bar{R}_j$ the mean of the ranks in group j , and $\bar{R}$ the mean of all ranks,

$$\bar{R} = \frac{1}{n}(1 + 2 + \dots + n) = \frac{1}{n} \frac{1}{2} n(n+1) = \frac{n+1}{2}.$$

- The KW teststatistic is given by

$$KW = \frac{12}{n(n+1)} \sum_{j=1}^g n_j \left(\bar{R}_j - \frac{n+1}{2} \right)^2.$$

- The factor $\frac{12}{n(n+1)}$ is used so that KW has a simple asymptotic null distribution. In particular under H_0 , given that $\min(n_1, \dots, n_g) \rightarrow \infty$,

$$KW \rightarrow \chi_{g-1}^2.$$

- The exact KW-test can be executed by calculating the permutation null distribution (that only depends on $n_1, \dots, n_g$) to test

$$H_0 : f_1 = \dots = f_g \text{ vs } H_1 : \text{at least two means are different.}$$

- In order to allow H_1 to be formulated in terms of means, the assumption of locations shift should be valid.
- For DMH example this is not the case.
- If location-shift is invalid, we have to formulate H_1 in terms of probabilistic indices:

$$H_0 : f_1 = \dots = f_g \text{ vs } H_1 : \exists j, k \in \{1, \dots, g\} : P[Y_j \geq Y_k] \neq 0.5$$

DNA Damage Example

```
kruskal.test(length ~ dose, data = dna)
```

Kruskal-Wallis rank sum test

data: length by dose

Kruskal-Wallis chi-squared = 14, df = 3, p-value = 0.002905

- On the 5% level of significance we can reject the null hypothesis.
- R-functie `kruskal.test` only returns the asymptotic approximation for p -values.

- With only 6 observaties per groep, this is not a good approximation of the p -value
- With the `coin` R package we can calculate the exacte p -value

```
library(coin)
kwPerm <- kruskal_test(length ~ dose,
  data = dna,
  distribution = approximate(B = 100000)
)
kwPerm
```

Approximative Kruskal-Wallis Test

data: length by dose (0, 1.25, 2.5, 5)
 chi-squared = 14, p-value = 0.00043

- We conclude that the difference in the distribution of the DNA damages due to the DMH dose is extremely significantly different.
- Posthoc analysis with WMW tests.

```
pairwise.wilcox.test(dna$length, dna$dose)
```

Pairwise comparisons using Wilcoxon rank sum exact test

data: dna\$length and dna\$dose

	0	1.25	2.5
1.25	0.013	-	-
2.5	0.013	0.818	-
5	0.013	0.721	0.788

P value adjustment method: holm

- All DMH behandelingen are significantly different from the control.
- The DMH are not significantly different from one another.
- U1 does not occur in the `pairwise.wilcox.test` output. Point estimate on probability on higher DNA-damage?

```

nGroup <- table(dna$dose)
probInd <- combn(levels(dna$dose), 2, function(x) {
  test <- wilcox.test(length ~ dose, subset(dna, dose %in% x))
  return(test$statistic / prod(nGroup[x]))
})
names(probInd) <- combn(levels(dna$dose), 2, paste, collapse = "vs")
probInd

```

```

      0vs1.25      0vs2.5      0vs5 1.25vs2.5  1.25vs5  2.5vs5
0.0000000 0.0000000 0.0000000 0.4444444 0.2777778 0.3333333

```

Because there are doubts on the location-shift model we draw our conclusions in terms of the probabilistic index.

Conclusion

- There is an extremely significant difference in in the distribution of the DNA-damage measurements due to the treatment with DMH ($p < 0.001$ KW-test).
- DNA-damage is more likely upon DMH treatment than in the control treatment (all $p=0.013$, WMW-testen).
- The probability on higher DNA-damage upon exposure to DMH is 100% (Calculation of a CI on the probabilistic index is beyond the scope of the course)
- There are no significant differences in the distributions of the comit-lengths among the treatment with the different DMH concentrations ($p = 0.72-0.82$).
- DMH shows already genotoxic effects at low dose.
- (Alle paarswise tests are gecorrected for multiple testing using Holm's methode).